

How few components suffice to compute one output token? (67M Pile target, C=256)

Per input, per target token: reward = logit of the gold next token; the sensor scores every component; the top-k are KEPT and all other shares zeroed.
Because the softpart shares sum to 1 at each weight entry, k=C restores the model exactly — every curve returns to the unablated CE, max |error| 2.0e-04.
5 seeds × 48 held-out target tokens. The ranking CROSSES OVER with budget: GIM is best below k≈16, but the IG family wins decisively from k≈32 and reaches the unablated CE with 64 of 256 components, where GIM needs ~205 and EAP never gets there. MIB's CPR/CMD is a different metric on different models and tasks.

